

Derivation: Maximum Efficiency of Slotted Aloha

Problem

For N nodes, each transmitting in a slot with probability p , the probability that *exactly one* node transmits (the efficiency) is

$$E(p) = Np(1-p)^{N-1}.$$

Find p^* that maximizes $E(p)$, compute the maximum $E(p^*)$, and show the limit as $N \rightarrow \infty$ gives $1/e$.

First derivative and optimum

Differentiate $E(p)$ with respect to p . Using product and chain rules,

$$\begin{aligned} E(p) &= Np(1-p)^{N-1}, \\ \frac{dE}{dp} &= N \left[(1-p)^{N-1} + p(N-1)(1-p)^{N-2}(-1) \right] \\ &= N(1-p)^{N-2} \left[(1-p) - p(N-1) \right] \\ &= N(1-p)^{N-2} (1-pN). \end{aligned}$$

Set $\frac{dE}{dp} = 0$. Since $(1-p)^{N-2} = 0$ corresponds to the trivial boundary $p = 1$, the interior critical point satisfies

$$1 - pN = 0 \quad \implies \quad \boxed{p^* = \frac{1}{N}}.$$

Second derivative (check it's a maximum) - Not Needed

Differentiate again (we show the sign at p^*):

$$\begin{aligned} \frac{d^2E}{dp^2} &= N \frac{d}{dp} \left[(1-p)^{N-2} (1-pN) \right] \\ &= N \left[-(N-2)(1-p)^{N-3} (1-pN) + (1-p)^{N-2} (-N) \right] \\ &= N(1-p)^{N-3} \left[-(N-2)(1-pN) - N(1-p) \right]. \end{aligned}$$

After algebraic simplification one obtains

$$\frac{d^2E}{dp^2} = N(1-p)^{N-3} (N-1)(Np-2).$$

Evaluate at $p^* = \frac{1}{N}$:

$$\left. \frac{d^2E}{dp^2} \right|_{p=1/N} = N \left(1 - \frac{1}{N} \right)^{N-3} (N-1)(1-2) = -N(N-1) \left(1 - \frac{1}{N} \right)^{N-3} < 0.$$

Thus $p^* = \frac{1}{N}$ is indeed a local maximum (and actually the global maximum on $0 < p < 1$).

Maximum efficiency

Substitute $p^* = 1/N$ into $E(p)$:

$$E(p^*) = N \cdot \frac{1}{N} \left(1 - \frac{1}{N}\right)^{N-1} = \left(1 - \frac{1}{N}\right)^{N-1}.$$

Limit as $N \rightarrow \infty$

We use the standard exponential limit. First recall

$$\lim_{N \rightarrow \infty} \left(1 - \frac{1}{N}\right)^N = e^{-1}.$$

Then

$$\lim_{N \rightarrow \infty} E(p^*) = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N}\right)^{N-1} = \lim_{N \rightarrow \infty} \frac{\left(1 - \frac{1}{N}\right)^N}{1 - \frac{1}{N}} = \frac{1/e}{1} = \frac{1}{e}.$$

(Equivalently, take natural logs: $\ln \left(\left(1 - \frac{1}{N}\right)^N \right) = N \ln \left(1 - \frac{1}{N}\right) \rightarrow -1$ by the Taylor expansion $\ln(1 - x) = -x + o(x)$ as $x \rightarrow 0$.)

Final boxed result

$p^* = \frac{1}{N}, \quad \max_p E(p) = \left(1 - \frac{1}{N}\right)^{N-1}, \quad \lim_{N \rightarrow \infty} \max_p E(p) = \frac{1}{e} \approx 0.3679$
